

Physics 4140H: Assignment II

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Problem 1. By using the Euler equations with an auxiliary condition imposed, find the shortest path between any two points on the surface of a sphere.

Solution 1. We impose the constraint that $g(x) = x^2 + y^2 + z^2 - R^2 = 0$, i.e. that we are on the surface of a sphere of radius R . Euler's equations are

$$\begin{aligned}\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) + \lambda(x) \frac{\partial g}{\partial y} &= 0 \\ \frac{\partial f}{\partial z} - \frac{d}{dx} \left(\frac{\partial f}{\partial z'} \right) + \lambda(x) \frac{\partial g}{\partial z} &= 0\end{aligned}$$

where f takes the usual form for distance-measure, $f = \sqrt{1 + (y')^2 + (z')^2}$. Hence, the first two derivatives are zero and

$$f_{y'} = \frac{y'}{\sqrt{1 + (y')^2 + (z')^2}}$$

and

$$f_{z'} = \frac{z'}{\sqrt{1 + (y')^2 + (z')^2}}$$

and $g_y = 2y$ and $g_z = 2z$. So, Euler's equations become

$$\begin{aligned}\frac{d}{dx} \left(\frac{y'}{\sqrt{1 + (y')^2 + (z')^2}} \right) &= 2y\lambda(x) \\ \frac{d}{dx} \left(\frac{z'}{\sqrt{1 + (y')^2 + (z')^2}} \right) &= 2z\lambda(x)\end{aligned}$$

which we can rewrite as a single equation to arrive at

$$\frac{1}{y} \frac{d}{dx} \left(\frac{y'}{\sqrt{1 + (y')^2 + (z')^2}} \right) = \frac{1}{z} \frac{d}{dx} \left(\frac{z'}{\sqrt{1 + (y')^2 + (z')^2}} \right)$$

this is tantalizingly close to something nice but I can't think of a way to incorporate the $1/y$ and $1/z$ so I could get rid of the derivatives, so we'll take the derivatives in full detail. So,

$$\frac{d}{dx} \left(\frac{y'}{\sqrt{1 + (y')^2 + (z')^2}} \right) = \frac{y'' \sqrt{1 + (y')^2 + (z')^2} - \frac{y'(y'y'' + z'z'')}{\sqrt{1 + (y')^2 + (z')^2}}}{1 + (y')^2 + (z')^2}$$

and similarly

$$\frac{d}{dx} \left(\frac{z'}{\sqrt{1 + (y')^2 + (z')^2}} \right) = \frac{z'' \sqrt{1 + (y')^2 + (z')^2} - \frac{z'(y'y'' + z'z'')}{\sqrt{1 + (y')^2 + (z')^2}}}{1 + (y')^2 + (z')^2}$$

and so the equality becomes, multiplying by yz on either side and cancelling the common denominator,

$$zy'' \sqrt{1 + (y')^2 + (z')^2} - \frac{zy'(y'y'' + z'z'')}{\sqrt{1 + (y')^2 + (z')^2}} = yz'' \sqrt{1 + (y')^2 + (z')^2} - \frac{yz'(y'y'' + z'z'')}{\sqrt{1 + (y')^2 + (z')^2}}$$

multiplying by the radical

$$zy'' + \cancel{z(y')^2 y''} + z(z')^2 y'' - \cancel{z(y')^2 y''} - zz' z'' y' = z'' y + y(y')^2 z'' + \cancel{(z')^2 z'' y} - z' y y' y'' - \cancel{(z')^2 z'' y}$$

so

$$zy'' + z(z')^2 y'' - zz' z'' y' = z'' y + y(y')^2 z'' - z' y y' y''$$

now we want to obtain, using the hint, an expression involving x . The only way we can get that is by using the $g(x)$ condition. Note that

$$\frac{dg}{dx} = x + yy' + zz' = 0 \text{ or } -x = yy' + zz'$$

so let's factor the equality to try to obtain $yy' + zz'$. Rearranging,

$$zy'' + z(z')^2 y'' + z' y y' y'' = z'' y + y(y')^2 z'' + zz' z'' y'$$

and factoring out the second derivatives

$$(z + z(z')^2 + z' y y') y'' = (y + y(y')^2 + zz' y') z''$$

and pulling out the extra z' and y' terms from the second two terms in each set of parentheses,

$$(z + [zz' + yy'] z') y'' = (y + [yy' + zz'] y') z''$$

and so using the derivative of the constraint,

$$(z - xz') y'' = (y - xy') z''.$$

Yay! Now to show that the equation of a plane passing through the origin, $Ax + By = z$, is a solution here. Note that $z' = A + By'$ and $z'' = By''$. Then $y = \frac{1}{B}(z - Ax)$, $y' = \frac{1}{B}(z' - A)$ and $y'' = \frac{1}{B}z''$ so plugging these in,

$$(Ax + By - x(A + By')) \left(\frac{1}{B} z'' \right) = \left(\frac{1}{B} (z - Ax) - x \left(\frac{1}{B} (z' - A) \right) \right) By''.$$

Multiplying things out

$$(\cancel{Ax} + By - \cancel{Ax} - Bxy') \left(\frac{1}{B} z'' \right) = \left(\frac{1}{B} z - \cancel{\frac{A}{B}x} - x \frac{1}{B} z' + \cancel{\frac{A}{B}x} \right) By''$$

and clearly by cancelling the remaining B factors

$$(y - xy') z'' = (z - xz') y''$$